Duodenum

The duodenum is the first part of your small intestine. Tubes entering the duodenum carry chemicals important for digestion from the liver and pancreas.

The muscular walls of the duodenum continue to churn the food making sure the digestive chemicals are mixed thoroughly with the food.

Oesophagus

The oesophagus is a long muscular tube that connects the mouth to the stomach. The muscles of the oesophagus push the ball of food down the tube by contracting behind the ball, then relaxing. This process is called **peristalsis**.

A flap of skin called the **epiglottis** closes the end of the windpipe as you swallow so the food does not enter your lungs.

A circle of muscle (called a **sphincter**) separates the oesophagus from the stomach. The sphincter closes after the food passes into the stomach to stop food

Stomach

Food is stored in the stomach for 1-6 hours. The muscular walls of the stomach contract and relax, churning up the food. This further digests the food mechanically as well as mixing it with **gastric juice**.

Gastric juice is made by special cells in the stomach wall and contains:

- **Hydrochloric acid** – a strong acid that kills bacteria that may have entered
- **Mucus** – creates a layer on the lining of the stomach so it does not digest itself
- **Pepsin** – an enzyme which helps break down large protein molecules and fats

Pancreas

The pancreas is not part of the digestive tract. It is a 15cm long organ that produces **pancreatic juice**, which is

Small Intestine

The part of the small intestine after the duodenum is the longest part of the digestive tract. It can be up to six metres long. The small intestine gets its name because of its diameter, which is only 3 cm.

Digested food is absorbed into the body in this part of the intestine. On the inner lining of the intestine are **villi** (singular: villus). These are microscopic fingers that greatly increase the surface area of the intestine wall and through which nutrients can pass.

Nutrients that pass from the small intestine are:

- **Glucose** from carbohydrates
- **Amino acids** from proteins

Large Intestine

The large intestine is the final part of the digestive tract. It is 1.5m long, and is wider than the small intestine, being about 6-7 cm in diameter.

In the large intestine, **water** is reabsorbed from what is left of the food, along with any remaining nutrients. The waste forms lumps of **faeces** which are later expelled from the body though the **anus** – a

Mouth

Digestion begins in the mouth, where teeth bite, cut and grind food into smaller pieces. This is **mechanical** digestion.

The food is mixed with saliva which contains an enzyme, called **amylase**, which starts to break down starch. This is the beginning of **chemical** digestion.

The tongue rolls the food into a ball, called a **bolus**, which is pushed down into the

Gall Bladder

The gall bladder is a muscular sack that stores up to 50mL of bile produced by the

Liver

The liver is the largest internal organ in the human body. It performs over 500 different chemical processes. It is a **vital organ**, meaning you cannot survive without it.

The process in the liver that is important to digestion is the production of **bile**, which is secreted into the duodenum. Bile is a greenish liquid responsible for the digestion of fats, by breaking them down into smaller pieces.

Some other functions of the liver include:

- Converting excess glucose to glycogen for storage
- Storage of vitamins and minerals
- Production of blood clotting

Salivary Glands

The salivary glands produce about 1L of saliva each day. It is a mixture of water, mucus and the enzyme **amylase**. Amylase begins breaking down starch into the simple sugar glucose.

Pancreas

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Pancreatic juice contains

- More enzymes to help digest carbohydrates, fats and proteins
- A basic solution to neutralise the acid in the chyme leaving the stomach.

The pancreas also produces the hormone **insulin**. Insulin is a chemical that controls

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